ARNOLD LOPEZ

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PROFESSIONAL EXPERIENCE

Interactive Simulation Programmer | Engineering & Computer Simulations **Jun 2022 – Aug 2025**
Ballston Spa, NY | Navy reactor-servicing virtual training

- Designed, implemented, and tested core systems for three major Unity-based virtual training products used to teach procedural equipment operations.
- Developed a GPU-based rope-physics system, improving performance stability and increasing prototype frame rate from approximately 90 to 120 FPS.
- Implemented networked VR tool and equipment interactions with Unity, Mirror Networking, and XR Interaction Toolkit for multi-user training scenarios.
- Investigated Unity profiling automation for Azure DevOps CI pipelines and maintained reusable packages and shared systems consumed by multiple projects.
- Built and deployed a standalone Python tool that automated cybersecurity STIG configuration work, reducing a repetitive compliance workflow.

Graphics Software Engineer | Leidos **May 2018 – Jul 2021**
Rockville, MD | Real-time periscope simulation

- Designed and implemented an OpenGL compute-shader particle system and standalone node-based particle editor using C++, Qt, CMake, and CPack.
- Diagnosed rendering defects across atmospheric sky, ocean simulation, post-processing, and particle pipelines, improving visual fidelity and runtime stability.
- Integrated SteamVR and OpenVR headset tracking into networked systems to provide synchronized VR interaction and visualization.
- Collaborated with artists and engineers to translate visual requirements into maintainable real-time rendering and authoring-tool workflows.

iOS Application Developer | Mobile Consulting Solutions **Jun 2017 – Jan 2018**
Atlanta, GA | Commercial aviation training application for Gulfstream pilots

- Implemented production Swift UI features and maintained airport and runway datasets using SQLite and Jeppesen source data.
- Collaborated with engineers, designers, and QA in an agile team to test, debug, and deliver reliable releases for an operational aviation-training application.

Game Programmer | Incredible Technologies **2016**
Vernon Hills, IL | Golden Tee Golf Mobile

- Contributed gameplay logic, UI features, scripting, and testing to a shipped commercial mobile game.

SELECTED PROJECTS

ForgeSim | *Modern C++ real-time simulation and game-engine framework*

- Building a target-based CMake architecture with a fixed-timestep runtime, OpenGL renderer, editor core, Dear ImGui frontend, and headless validation.
- Maintaining strict separation between engine/simulation services and UI code to support future frontends and adapters without rewriting domain models.

Terrasweeper | *Unity / C# Mobile Puzzle Game | 2016*

- Designed, developed, and released an original terrain-based reinterpretation of Minesweeper on Google Play, featuring three difficulty levels and Google Play achievements.

Dungeon Generator Remaster | *Unity / C# procedural-generation portfolio project*

- Developed procedural dungeon-generation and gameplay systems, demonstrating reusable runtime architecture and content-generation workflows.

EDUCATION

B.S. Computer Game Development | DePaul University, Chicago, IL **2015**

TECHNICAL SKILLS

Languages: C++, C#, Python, Swift, Lua
Engines & Simulation: Unity, Unreal Engine, XR Interaction Toolkit, Mirror Networking, OpenSceneGraph
Graphics & Profiling: OpenGL, GLSL, HLSL, compute shaders, NVIDIA Nsight Graphics
Tools & Delivery: Qt, CMake, CPack, Git, Azure DevOps, Unity Test Framework, SQLite